

Stat 134: Concepts of Probability, Fall 2026

Team:

Professors:

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GSIs:

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Adjunct (Stat 198) instructor: Mike Leong mleong@berkeley.edu.

OFFICE HOURS (in Student Learning Center Atrium starting week 2):

Adam: **W 11-1pm**

Shirshendu: **F 2-4pm**

GSI and tutors: see OH link on bCourses/pages

Class Procedure

Lectures: MWF 1-2 pm Undergraduate Academic Building 100

Discussion: Tuesday, Thursday. Location and time vary by section:

1. Section 101 from 5:00 - 6:00 in B51-Hildebrand
2. Section 102 from 9:00 - 10:00 in 20-Wheeler
3. Section 103 from 9:00 - 10:00 in 88 Dwinelle
4. Section 104 from 10:00 - 11:00 in B56-Hildebrand
5. Section 105 from 11:00 - 12:00 in 3111-Etcheverry
6. Section 106 from 12:00 - 1:00 in 3111-Etcheverry
7. Section 107 from 1:00 - 2:00 in B1-Hearst Field Annex

8. Section 108 from 1:00 - 2:00 in 242-Hearst Gym
9. Section 109 from 1:00 - 2:00 in 120-Wheeler
10. Section 110 from 5:00 - 6:00 in 242-Hearst Gym
11. Section 111 from 4:00 - 5:00 in 242-Hearst Gym
12. Section 112 from 4:00 - 5:00 in B51-Hildebrand

Textbook: *Probability* by Jim Pitman. Students can view and download the text online at **SpringerLink** (campus network access required). An affordable paperback copy may also be purchased through this link.

Course Websites:

Announcements will be posted on bCourses and Ed. Please submit daily homework to Gradescope.

Contact:

Email: stat134question@lists.berkeley.edu

Please always “reply-all” when replying to stat134question@lists.berkeley.edu so the communication goes to everyone on our support team.

Don't send email through Canvas.

Class Outline:

Course Description:

An introduction to probability, emphasizing concepts and applications without coding components. Possible topics include: Axioms of probability. Discrete and continuous random variables. Distributions and relations between random variables. Independence and dependence. Joint, marginal, and conditional random variables. Bayes rule. Expectation and variance. Covariance and correlation. Concentration inequalities, tail probabilities and introductory asymptotics. Law of large numbers. Central limit theorem. Change of density and sampling.

Schedule: See the end of the syllabus.

Prerequisites:

Fluency in calculus: derivatives and integrals of functions in one and two variables; computations of sums and infinite series. *Clear logical reasoning and problem-solving skills.*

Please take the diagnostic test (due Friday August 28) so we know if you have sufficient background for Stat 134. Graders will grade your diagnostic for correctness, but it won't count towards your grade.

On the page for Data 140, Professor Adhikari has a "[Math Prerequisites](#)" page. Please look at it (skip linear algebra). An excellent resource for reviewing the prerequisites for Stat 134 is Anderson's Introduction to Probability (Appendix A,B,C,D), available online through the library.

Success and Challenges:

Our goal is to provide a welcoming, collaborative learning environment for exploration and growth. Our job is to serve your development first. We will work to give you the tools to succeed. Success in a course is relative to your desired outcomes, so you should think about your learning objectives. A grade is not a learning objective.

Importantly, your grade is not your worth. We do not judge students' character from their class performance. Rather, productive struggle is the best hallmark of learning. So, if you are struggling, for any reason, come talk to us. We will do our best to provide clear, achievable routes to success. The whole course staff are here to help and support you.

Policies:

Lecture Attendance:

Attendance is expected and encouraged. Attending lecture gives us the opportunity to know you better and for this reason we only write letters of recommendation for students that attend lecture regularly. Lecture recordings will only be available on advance request or under extenuating circumstances.

Recording Policy:

We believe strongly that the virtue of lecture is the live, regular, segmented format. We will do our best to make live lecture worth attending. Lecture will be recorded automatically through course capture. Recordings will be completed after class and will not be released live. Recordings will not be posted but will be shared given advance notice for extenuating circumstances. Please contact course staff as early as possible if you anticipate needing a recording for a legitimate class conflict or miss lecture due to illness or unavoidable circumstances. Foreseeable scheduling conflicts or time mismanagement are not valid conflicts. Professional obligations are (e.g. conference travel, etc.). *All students will get two "free" recordings on request - no rationale needed. Requests made after class time will be counted as a "free" recording. All other requests need a rationale.*

Lectures missed due to late enrollment will be provided on request, without counting against your 2 free recordings.

To request a recording, submit the recording request form:

<https://docs.google.com/forms/d/e/1FAIpQLSdvQ3VRCY0iEtwxS0KL3DKlWRpActAIYA1rJEtOZLQ7IU6ESA/viewform?usp=sharing&ouid=117659300599081425344>

We will do our best to respond quickly to requests for lecture recordings - but do not promise to post in less than 24 hrs after a request (see the communication policy above), especially if requests are made outside regular working hours. *Barring exceptional circumstances, we will not provide recordings requested outside the week of the recording.* So, if for any reason, you anticipate missing a class, or are forced to miss a class, please contact us early.

We also strongly encourage you to make friends in class and to share notes. Most of the lecture content is also available in course reading. If you do miss a class, and are unsure what to read, ask the course staff.

Discussion (Section) Attendance:

Discussion attendance is expected. Discussion worksheets will be collected to track course participation and will be graded for completion and attendance. You must attend your assigned section. Discussion is an important part of class where students get practice working on problems related to the lecture and homework.

Late Work Policy:

We will not accept late work. Note that, by design, the grading policy allows a certain number of HW and discussion drops without penalty. If you join the class late you must use your drops up front.

LLM and AI use:

We can't police the use of AI for HW, so we won't. Moreover, some uses of AI are productive. So you are welcome to use it. However, we strongly advise against using it as a crutch, and suggest you attempt problems on your own first, or with peers. *You cannot learn to problem solve by reading solutions.* It is your responsibility to take ownership of your practice and study. If you do not master the problems presented in the HW, you will not be prepared for the quizzes. If you do, then you will be well prepared and will not need to spend much additional time studying. If you find you are relying on AI, or complete the HW but struggle on quizzes and the midterm come talk to us so we can help you work out a plan to practice more productively

Activities and Assignments:

The following course components will be designed to 1. give an introduction to the material, 2. provide hands-on practice, 3. evaluate proficiency, 4. evaluate mastery.

1. Reading: Textbook readings will be suggested in advance. We strongly suggest that you do the reading before coming to lecture and before attempting practice problems.

2.a Discussion Problems: Each discussion section will be used to solve practice problems. Discussion sessions will mix guided practice with group work. A discussion worksheet will be collected by your GSI at the end of section and checked for

completion. They will be returned in the next session. Late discussion worksheets will not be accepted.

2.b Homework Assignments: Daily assignments are posted on bCourses after each class. They are due on the date given on Gradescope . **It is your responsibility to upload HW properly. Always double check that you submitted properly well before the deadline. You cannot resubmit after the deadline.**

Daily assignments will be graded on a 0/1 scale. Each assignment will consist of 3-5 problems. A good attempt on **all** problems will receive the score of 1; anything else will get 0. Course readers will determine what constitutes a "good attempt." We will be looking for reasoning and detailed work shown – it is assumed that you will show work or provide reasoning whether or not the question asks for it. We provide minimal practice and highly recommended problems that you can test yourself on.

3. Quizzes: You will take 6 quizzes over the semester. Quizzes will be held through the Computer Based Testing Facility (CBTF) <https://rtl.berkeley.edu/services-programs/computer-based-testing-facility-cbtf>. Note that DSP students will also take it in the CBTF and will get their extended time. **You can see the dates for all quizzes and retakes on PrairieTest <https://us.prairietest.com/>.** You will have to schedule your quiz during the given dates at the CBTF. We do not schedule your quiz for you. Note that DSP students will also take it in the CBTF and will get their extended time. Quizzes will consist of multiple choice or select all that apply or numerical fill-in-the-blank questions. They are meant to test your understanding of the material. You are allowed a cheat sheet for quizzes, but you must upload your cheat sheet to PrairieTest before the quiz/retake. CBTF provides a built-in calculator for all quizzes/retakes. Quizzes will be graded on a complete/incomplete basis at an 80% threshold. Quiz grades will not be curved. Grades between a 60% and an 80% will count as half complete.

Quiz Retake Policy: You will have the opportunity to **retake a total of three quizzes**. Due to space limitations the CBTF only allows 40% of the class to retake each quiz—hence if you get a score of >80% a quiz (i.e. complete the quiz) do not schedule a retake. *Retakes will not be offered for missed quizzes unless we are notified in advance and excuse your absence.*

4. Exams: As below. Written exams (1 midterm, 1 cumulative final) will evaluate mastery of the course material.

Midterm: The midterm will be administered outside of class on Monday October 19 at 7pm location TBA. Students will be given 110 minutes. There is no lecture the day of the midterm.

Final: The final exam will be a cumulative written exam held on Wednesday December 16. 7-10pm (location TBA).

Clobber Policy: If we are given advance notice with appropriate documentation that you must miss the midterm, then we may grant a clobber. The clobber policy

will replace your midterm with the average of your quiz scores on the matching units (e.g. midterm = average of quizzes 1 to 3). The clobbered midterm score will be automatically weighted 15% in your exam average, with 85% to your final exam.

Grades:

Grade Breakdown:

Per the division of assignment roles outlined above, your grade will not be based on a weighted average of components. Instead, you need to participate (i.e., do the work), demonstrate proficiency across most topics (i.e., perform sufficiently well on most quizzes), and show mastery in the exams. Grades will be computed from three categories:

Participation (Discussion + Homework): Your participation average is the average of your HW and discussion worksheet averages (weighted 50% HWs, 50% discussion worksheets). Note: There are 32 HW and 25 discussion worksheets. You may drop 6 HW and 5 discussion worksheets without any penalty to your grade. **If you add the class late you must use up your drops for any missed work.**

Proficiency (quizzes): There are a total of 6 quizzes graded I/H/C (incomplete/half-complete/complete). Any score at or above 80% (inclusive) is considered completion. Any score at or above 60% but beneath 80% is a half completion. You may retake all quizzes according to the retake policy specified above. Quiz completion will be assigned based on the better of your two attempts.

Mastery (exams): An average of your midterm and final exam.

Final Grade:

You will pass the class unless any of the following conditions happen:

- a. Your participation average is less than 45%
- b. You complete fewer than 1.5 quizzes
- c. Your final exam absolute score is less than 25%

Your mastery score will be calculated as 35% Midterm + 65% Final, and your mastery rank will be based on your mastery score percentile. Those in the top 30% of the class get some kind of A, the next 30% get some kind of B, and next 40% get some kind of C for your mastery grade.

Your final grade (assuming you pass) will be the mastery proficiency rank capped by the intersection of the participation and proficiency intervals, calculated as follows (see [this calculator](#) to play around interactively):

1. Participation interval: depends on the range of your participation score, which will be the average between your homework and your section participation percentages.
 1. [0%, 45%) -> Fail the class
 2. [45%, 60%) -> from F to B
 3. [60%, 70%) -> from C- to B+
 4. [70%, 80%) -> from C to A
 5. [80%, 100%] -> from C+ to A+
2. Proficiency interval: depends on the range of your total quiz score:
 1. 0-1 completes -> Fail the class
 2. 1.5-2 completes -> from C- to B+
 3. 2.5-3 completes -> from C to A-
 4. 3.5-4 completes -> from C+ to A
 5. 4.5-5 completes -> from B- to A+
 6. 5.5-6 completes -> from B to A+

Example scenarios:

Assume below you do better than 25% absolute score on the final exam.

- If your completion average for HW and discussion worksheets is 60% and you complete 1.5 quizzes, then your grade will be between a C- and a B+ depending on your mastery average percentile rank. If for example your mastery average percentile rank in the class is 70% (some kind of A) your course grade is a B+ since it is capped at a B+.
- For it to be possible to get an A in the class you need to complete at least 3.5 quizzes and have a completion average of 70% on HW and discussion worksheets.
- If your completion average on HW and discussion worksheets is 80% and you complete 5.5 quizzes, then your grade will be between a B and an A+ depending on your mastery average percentile.

Rights, Integrity, Conflicts, Accommodations, Safety

Student and Instructor Rights: This course is an open discussion space. All members (students and instructor) have the right to academic freedom in all course spaces (lecture, Ed, etc). Here academic freedom means freedom of expression and opinion. However, freedom of expression does not grant a right to demeaning or discriminatory behavior. As a formal space, the instructor reserves the right to maintain community standards of mutual respect and inclusivity. The same standards applied to students extend to the instructor. All materials developed by the instructor for use in the course are the

intellectual property of the instructor and will not be distributed beyond the course without permission.

Honor code: Please see the [Berkeley Honor Code](#).

DSP Accommodations: Please see the "[Statement of Accommodations](#)" page. If you have an accommodation allowing extensions on assignments, you must contact me before the assignment is due to receive the extension, unless you are prevented from requesting the extension due to a DSP related reason. If you have testing accommodations, you may request to take the quizzes through DSP proctoring. To do so, you must schedule your proctoring time with DSP in advance. It is your responsibility to do so. Otherwise, all other quizzes will be taken in class at the end of class (allowing 15 minutes). Specific quiz dates are listed in the calendar below.

Academic Integrity and Scheduling Conflicts: Please see the [Berkeley academic integrity and conflicts page](#). Note: use of generative AI to generate text will be considered plagiarism. Regarding scheduling conflicts: please contact us early the course email stat134question@lists.berkeley.edu.

Safety: Please see the [Berkeley Office of Emergency Management](#) for earthquake procedures.

	Monday	Wednesday	Friday
24-Aug		1.1-1.3 (lab 1)	1.4-1.5 (diag-due) (lec 2)
31-Aug	1.5, 2.1 (HW 1)	2.1, 2.2 (HW 2) (3)	2.4 (skip 2.3), 2.5 (HW 3) (5)
7-Sep	holiday	2.5, 3.1 (HW 4) (4)	3.1, 3.2 (HW 5). (7)
14-Sep	3.2 (HW 6) Q1	3.2, 3.3 (HW 7) (6)	3.3 (HW 8) (10) Q1 retake
21-Sep	3.6 (HW 9)	3.6, 3.4 (HW 10) (8) Q2	3.5 (HW 11) (13)
28-Sep	4.1, 4.2 (HW 12) Q2 retake	4.2 (HW 13) (10)	4.2 (HW 14) (16)
5-Oct	(skip 4.3) 4.4 (HW 15)	MGF (12) (HW 16)	MGF (HW 17) (19)
12-Oct	review (HW 18)	review (14) Q3	review (22)
19-Oct	midterm	4.5 (15)	4.5 (HW 23) (24)
26-Oct	5.1, 5.2 (HW 24) Q3 retake	5.1, 5.2 (HW 25) (17)	5.3 (HW 26) (27)
2-Nov	5.3 (HW 27) Q4	5.4 (HW 28) (19)	5.4 (HW 29) (30)
9-Nov	6.1, 6.2 (HW 30)	holiday	6.3 (HW 31) (32) Q4 retake
16-Nov	6.3 (HW 32) Q5	6.4 (HW 33) (22) Q5 retake	6.4 (HW 34) (35)
23-Nov	6.5 (HW 35)	holiday	holiday
30-Nov	6.5 (HW 36). Q6	review (HW 37) (25)	off (39)
RRR week	RRR review. Q6 retake	off	RRR review
14-Dec		Final Wed 7:00- 10:00pm	